

SWIGGY VS ZOMATO: RESTAURANT PERFORMANCE AND CUSTOMER INSIGHTS ANALYSIS



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CERTIFICATION

DECLARATION

I hereby certify that the work which is being presented in the project entitled “**SWIGGY VS ZOMATO: RESTAURANT PERFORMANCE AND CUSTOMER INSIGHTS ANALYSIS**” by Parneet Kaur in partial fulfillment of requirements for the award of degree of **B.Tech** submitted in the department of **Computer Science and Engineering** at University Institute Of Engineering and Technology under **Sant Baba Bhag Singh University, Jalandhar** is an authentic record of my own work carried out during a period from *June to July 2026*, under the supervision of **Mr. Harjinder Kumar in O7 Services**. The matter presented in this project has not been submitted by me in any other University / Institute for the award of B.Tech degree.

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This is to certify that the above statement made by the candidate is correct to the best of my / our knowledge.

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ABSTRACT

In India's expanding food delivery ecosystem, food tech aggregators **Swiggy** and **Zomato** command significant market share across urban hubs. Understanding key performance metrics—such as customer rating dynamics, order fulfillment times, platform commission fees, price category distribution, and monthly net profitability—is critical for restaurant partners, investors, and industry analysts.

This project, titled "**Swiggy vs Zomato: Restaurant Performance and Customer Insights Analysis**", provides an end-to-end data analytics and interactive web solution analyzing **2,500 unique restaurant listings** across **10 major Indian metropolitan cities** (Mumbai, Bangalore, Delhi, Pune, Hyderabad, Chennai, Kolkata, Ahmedabad, Jaipur, and Lucknow). The dataset evaluates both aggregators across 41 operational attributes.

Using Python's data science ecosystem—**NumPy** for numerical processing, **Pandas** for dataset manipulation and cleaning, **Matplotlib** and **Plotly Express** for data graphics, and **Streamlit** for dashboard application engineering—the platform processes multi-attribute data into clear visual insights. Key analytical findings include:

- **Platform Dominance:** Out of 2,500 evaluated restaurants, Zomato outperforms Swiggy in **1,273 listings (50.92%)**, while Swiggy leads in **1,195 listings (47.80%)**, and **32 listings (1.28%)** reflect balanced performance across both platforms.
- **Customer Satisfaction:** Zomato holds an average rating of **4.01 / 5.0**, whereas Swiggy averages **3.99 / 5.0**.
- **Delivery Performance:** Zomato logs an average delivery time of **39.57 minutes**, compared to Swiggy's **39.67 minutes**.
- **Revenue Performance:** Total estimated monthly revenue generated across all listings reaches **₹32.56 Crore** on Zomato and **₹32.45 Crore** on Swiggy.

This report presents the end-to-end technical methodology, software requirement specifications (SRS), data flow diagrams (DFD), dataset dictionary, implementation code architecture, visualization snapshots, and future enhancement scopes.



O7 Services is ISO 9001:2015 Certified company founded in 2015. They specialise in a variety of services including Web Development, Mobile Application Development, Custom Software Development, UI/UX Designing, Hosting services, Digital Marketing, Registration of Domain Names with various extensions, AMC & MMC Services, Bulk SMS and voice calls. O7 Services offers advanced IT solutions supporting the entire business cycle - from consulting to system development, deployment, quality assurance, and 24x7 support. With over 9 years of experience, the company aims to form long-lasting strategic partnerships with clients, offering affordable prices, timely delivery, and measurable business results. Their headquarter is located in Jalandhar with a branch office in Hoshiarpur.

Some of the products developed by O7 Services include Vehicle Tracking System, Invoice Software, School Management System, Hospital Management System, Parents - Teacher Communication App, Fee Management system, Task Management System, Online Food Ordering App, Security App, Admission system, Inventory Software, and Car Servicing App. Additionally, O7 Services provides training programs including 6 Weeks/6 Months Industrial Training, ProjectBased Training, Corporate Training, and Job-Oriented Courses Training, covering major IT trends such as Full Stack Development (MEAN/MERN), Flutter, Kotlin Android, Swift UI (iOS), Firebase, Python, Angular, React Js, Vue Js, Node Js, ASP.NET, .NET Core, PHP, Laravel, CodeIgniter, Software Testing, Cloud Computing, Blockchain, DevOps, Data Science, Artificial Intelligence, Machine Learning, UI/UX Designing, Digital Marketing, WordPress, Linux, CCNP, CCNA Security, Network Security, Cyber Security, MCSE, MCITP, Java, Spring, Hibernate, C/C++, Photoshop, Adobe Illustrator, Figma, CorelDraw and many more.

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DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Abbreviation	Full Form
IT	Information Technology
AI	Artificial Intelligence
ML	Machine Learning
EDA	Exploratory Data Analysis
UI / UX	User Interface / User Experience
DFD	Data Flow Diagram
ER Diagram	Entity-Relationship Diagram
SRS	Software Requirements Specification
KPI	Key Performance Indicator
QSR	Quick Service Restaurant
INR (₹)	Indian National Rupee
API	Application Programming Interface
CSV	Comma-Separated Values
XLSX	Excel Spreadsheet Format
RAM	Random Access Memory
UIET	University Institute of Engineering and Technology
SBBSU	Sant Baba Bhag Singh University

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INTRODUCTION TO ORGANIZATION

1.1 Industry Profile

Information Technology (IT) industry is one of the fastest-growing industries in the world and plays a significant role in digital transformation across various sectors. Businesses generate large volumes of data every day, making Data Science an essential field for analysing data and supporting informed decision-making.

The food delivery industry has experienced rapid growth due to increasing internet usage and smartphone adoption. Platforms such as **Swiggy** and **Zomato** provide online food ordering and delivery services, generating valuable data related to restaurant ratings, customer preferences, delivery performance, pricing, and revenue. This data helps businesses improve operational efficiency, understand customer behaviour, and identify market trends.

Data Science technologies such as **Python, NumPy, Pandas, Matplotlib, Seaborn, Plotly Express, and Streamlit** enable organizations to process, analyse, and visualize large datasets effectively. These tools are widely used in business analytics because they provide meaningful insights through statistical analysis and interactive dashboards.

Today, organizations in healthcare, banking, education, retail, transportation, and food delivery increasingly rely on Data Science to improve customer satisfaction, optimize operations, and make data-driven decisions.

1.2 General Information

O7 Services is an Information Technology company that provides software development, industrial training, and technical consulting services. The organization focuses on providing practical training to students through industry-oriented projects and modern software technologies.

The company offers training in several technical domains, including:

- Python Programming
- Data Science
- Artificial Intelligence
- Machine Learning
- Web Development

- Mobile Application Development
- Cloud Computing
- Software Testing

During the industrial training, students are introduced to real-world software development practices and modern analytical tools. They gain practical experience in programming, data analysis, visualization, and project development under the guidance of experienced professionals. For this project, O7 Services provided training in **Data Science with Python**, where the project "**Swiggy vs Zomato: Restaurant Performance and Customer Insights Analysis**" was developed. The project uses Python libraries to clean, analyse, and visualize restaurant data, enabling users to compare the performance of Swiggy and Zomato through an interactive Streamlit dashboard. The organization emphasizes practical learning, problem-solving, teamwork, and the application of technical knowledge to real-world business problems, helping students build skills required in the IT industry.

2.1 Introduction

The food delivery industry has grown significantly with the increasing use of smartphones and internet services. Online food delivery platforms provide customers with a convenient way to order food from a wide range of restaurants while allowing restaurant owners to reach a larger customer base. Among these platforms, **Swiggy** and **Zomato** are the two leading food delivery service providers in India.

These platforms generate a large amount of data related to restaurant ratings, delivery time, customer reviews, pricing, cuisine, and business performance. Analysing this data helps businesses understand customer preferences, evaluate restaurant performance, and make better business decisions. However, manual analysis of such large datasets is time-consuming and inefficient.

This project, "**Swiggy vs Zomato: Restaurant Performance and Customer Insights Analysis**" aims to analyse restaurant data and compare the performance of both platforms using Data Science techniques. The project applies Python libraries such as **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**, **Plotly Express**, and **Streamlit** to clean, process, analyse, and visualize the dataset. The dataset contains **2,500 restaurant records and 41 features**, including restaurant names, city, cuisine, ratings, delivery time, average cost, monthly revenue, estimated profit, and other business-related information.

The processed data is presented through an interactive Streamlit dashboard, allowing users to compare Swiggy and Zomato based on various business parameters. The dashboard displays visual reports and charts that make it easier to identify trends, compare performance, and generate useful insights.

The online food delivery industry has experienced rapid growth, making platforms such as Swiggy and Zomato an important part of the restaurant ecosystem. This project presents a comparative analysis of these two platforms using a structured restaurant dataset. The objective is to analyze and compare key business indicators, including restaurant ratings, pricing, delivery performance, revenue, monthly orders, cuisines, restaurant types, and city-wise distribution.

The project has been developed using Python, Pandas, Plotly, and Streamlit to transform raw restaurant data into meaningful visual insights. Interactive dashboards and visualizations allow users to explore the data from different perspectives by applying filters for city, cuisine, and

restaurant type. This helps in understanding platform performance and identifying trends across multiple business metrics.

The analysis focuses on comparing the operational performance of restaurants listed on both platforms rather than determining an overall winner. Various visualizations such as bar charts, line charts, scatter plots, treemaps, histograms, box plots, violin plots, heatmaps, and sunburst charts are used to examine relationships between pricing, delivery time, customer ratings, revenue, and restaurant characteristics.

Overall, the project demonstrates how data analytics and interactive dashboards can simplify restaurant performance analysis, making complex datasets easier to understand. The generated insights support comparative evaluation and provide a clear overview of restaurant performance across Swiggy and Zomato..

2.2 Objectives of the Project

The main objectives of the project are:

- To analyse restaurant data from Swiggy and Zomato.
- To clean and preprocess the dataset using Pandas.
- To perform numerical computations using NumPy.
- To compare customer ratings, delivery time, revenue, and profit.
- To create meaningful visualizations using Matplotlib, Seaborn, and Plotly Express.
- To develop an interactive dashboard using Streamlit.
- To generate business insights that support better decision-making.

2.3 Scope of the Project

The scope of the project includes analysing restaurant data to compare the performance of Swiggy and Zomato. The project focuses on:

- Data cleaning and preprocessing.
- Exploratory Data Analysis (EDA).
- Comparative analysis of both platforms.
- Revenue and profit analysis.
- Customer rating analysis.

- Delivery time comparison.
- Interactive dashboard development.
- Business insight generation.

The project is useful for restaurant owners, business analysts, students, and researchers interested in understanding trends in the online food delivery industry.

2.4 Technologies Used

Table 2.1: Technologies Used

Technology	Purpose
Python	Programming Language
NumPy	Numerical Computation
Pandas	Data Cleaning and Analysis
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Plotly Express	Interactive Charts
Streamlit	Interactive Dashboard
Jupyter Notebook	Data Analysis
Visual Studio Code	Application Development

2.5 Features of the Proposed System

The proposed system provides the following features:

- Upload and analyse restaurant datasets.
- Clean and preprocess data automatically.
- Compare Swiggy and Zomato performance.
- Analyse ratings, delivery time, and pricing.
- Generate interactive charts and graphs.
- Display Key Performance Indicators (KPIs).
- Filter data by city, cuisine, and restaurant type.

- Present analytical results through a Streamlit dashboard.

2.6 Benefits of the Project

The project offers several benefits:

- Reduces manual data analysis.
- Provides accurate statistical insights.
- Supports business decision-making.
- Helps compare restaurant performance across platforms.
- Improves understanding through interactive visualizations.
- Easy to use and scalable for larger datasets.

CHAPTER 3

PROJECT REVIEW

3.1 Feasibility Study

A feasibility study is conducted to determine whether the proposed project is practical, achievable, and beneficial. It evaluates the project from different perspectives before development begins.

3.1.1 Technical Feasibility

The proposed system is technically feasible as it is developed using Python and open-source libraries such as **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**, **Plotly Express**, and **Streamlit**. These technologies are compatible with standard computer systems and provide all the necessary tools for data analysis and dashboard development.

3.1.2 Economic Feasibility

The project is economically feasible because all the software, libraries, and development tools used are free and open-source. The project requires only a personal computer with Python installed, making the implementation cost very low.

3.1.3 Operational Feasibility

The system is user-friendly and can be easily operated by users with basic computer knowledge. The interactive Streamlit dashboard enables users to upload the dataset, apply filters, and view analytical reports without requiring technical expertise.

3.1.4 Schedule Feasibility

The project was completed within the industrial training period. The development activities, including data preprocessing, analysis, visualization, dashboard development, testing, and documentation, were completed according to the planned schedule.

3.2 Product Definition

The proposed product is an interactive **Data Science Dashboard** that analyses and compares restaurant data from **Swiggy** and **Zomato**. The system processes the dataset, performs statistical analysis, and presents meaningful insights through charts, graphs, and key performance indicators (KPIs).

The dashboard enables users to:

- Compare restaurant ratings on both platforms.
- Analyse delivery time and pricing.
- Evaluate revenue and estimated profit.
- Identify popular cuisines and restaurant categories.
- View city-wise restaurant performance.
- Generate business insights through interactive visualizations.

3.3 Problem Statement

The rapid growth of online food delivery platforms has resulted in the generation of large amounts of restaurant data. Manually analysing this data to compare restaurant performance, customer ratings, pricing, and delivery efficiency is time-consuming and difficult.

The proposed system addresses this problem by applying Data Science techniques to analyse restaurant data from Swiggy and Zomato. It provides meaningful insights through interactive dashboards and visual reports, enabling users to make informed business decisions quickly and efficiently.

3.4 Project Functionality

The major functions of the system are:

- Load restaurant dataset.
- Clean and preprocess data.
- Handle missing and duplicate values.
- Perform Exploratory Data Analysis (EDA).
- Compare Swiggy and Zomato.
- Generate interactive visualizations.
- Display business metrics and KPIs.
- Filter data based on city, cuisine, and platform.
- Present analytical reports through the Streamlit dashboard.

3.5 System Requirements

3.5.1 Hardware Requirements

Table 3.1: Hardware Requirements

Component	Specification
Processor	Intel Core i3 or above
RAM	4 GB (8 GB Recommended)
Storage	500 MB Free Space
Operating System	Windows 10/11

3.5.2 Software Requirements

Table 3.2: Software Requirements

Software	Purpose
Python	Programming Language
NumPy	Numerical Computation
Pandas	Data Analysis
Matplotlib	Data Visualization
Seaborn	Statistical Charts
Plotly Express	Interactive Charts
Streamlit	Dashboard Development
Jupyter Notebook	Data Analysis
Visual Studio Code	Application Development

3.6 Methodology

The project follows the **Data Science Life Cycle**, which consists of the following stages:

Step 1: Data Collection

The restaurant dataset is collected in CSV format containing restaurant and business-related information.

Step 2: Data Preprocessing

The dataset is cleaned by handling missing values, removing duplicate records, correcting data types, and preparing the data for analysis.

Step 3: Exploratory Data Analysis (EDA)

The processed data is analysed using statistical techniques to identify trends, relationships, and patterns.

Step 4: Data Visualization

Charts and graphs are created using Matplotlib, Seaborn, and Plotly Express to represent the analytical findings visually.

Step 5: Dashboard Development

A user-friendly dashboard is developed using Streamlit to display interactive charts, KPIs, and business insights.

Step 6: Testing and Validation

The dashboard is tested to ensure that all visualizations, filters, and calculations produce accurate results.

3.7 Acceptance Criteria

The project is considered successful if it satisfies the following conditions:

- The dataset is loaded successfully.
- Data preprocessing is completed without errors.
- Missing and duplicate values are handled correctly.
- Analytical reports are generated accurately.
- Interactive charts and dashboards function properly.
- Users can compare Swiggy and Zomato based on multiple parameters.
- The application runs smoothly and provides meaningful business insights.

Analysis:

The dataset was analyzed using descriptive statistics and interactive visualizations to compare the performance of Swiggy and Zomato across multiple business metrics. Key parameters such as restaurant ratings, pricing, delivery time, revenue, monthly orders, restaurant type, cuisine, and city-wise distribution were examined to identify patterns and performance differences.

The analysis indicates that restaurant performance is influenced by several factors rather than a single metric. Delivery time generally increases as the distance from the city center grows, while pricing and revenue vary across restaurant categories and cities. Customer ratings remain relatively consistent across most price categories, suggesting that higher pricing does not necessarily guarantee higher customer satisfaction.

Interactive filters further enabled comparisons based on city, cuisine, and restaurant type, making it easier to explore platform-specific trends and identify performance variations. Overall, the analysis provides meaningful insights into restaurant operations and highlights the usefulness of data visualization for comparing food delivery platforms.

This chapter presents the analysis of the **Swiggy vs Zomato Restaurant Dataset**. It explains the flow of data through the system, the overall project workflow, dataset processing, report format, data dictionary, and the functional and performance specifications of the proposed analytics dashboard. The analysis focuses on comparing restaurant performance across the Swiggy and Zomato platforms using Python-based data analytics techniques.

4.1 External Interface and Data Flow

The proposed system provides an interactive interface developed using **Streamlit**, enabling users to upload the restaurant dataset, analyse restaurant performance, compare platform statistics, and visualize trends through interactive charts.

Input Interface

- Upload Swiggy vs Zomato dataset (CSV format).

- Select filters such as city, cuisine, restaurant type, and platform.
- Choose different analysis options.
- View dataset summary and statistics.

Processing Interface

- Read the uploaded dataset.
- Handle missing values and duplicate records.
- Convert columns into appropriate data types.
- Perform Exploratory Data Analysis (EDA).
- Generate statistical summaries.
- Create visualizations using Python libraries.

Output Interface

- Display cleaned dataset.
- Show restaurant performance comparison.
- Present interactive graphs and dashboards.
- Generate business insights for both platforms.

4.1.1 Data Flow

The project follows the following sequence:

- Dataset Collection
- Data Loading
- Data Cleaning
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Restaurant Comparison Analysis
- Data Visualization
- Streamlit Dashboard

- Result Interpretation
- Report Generation

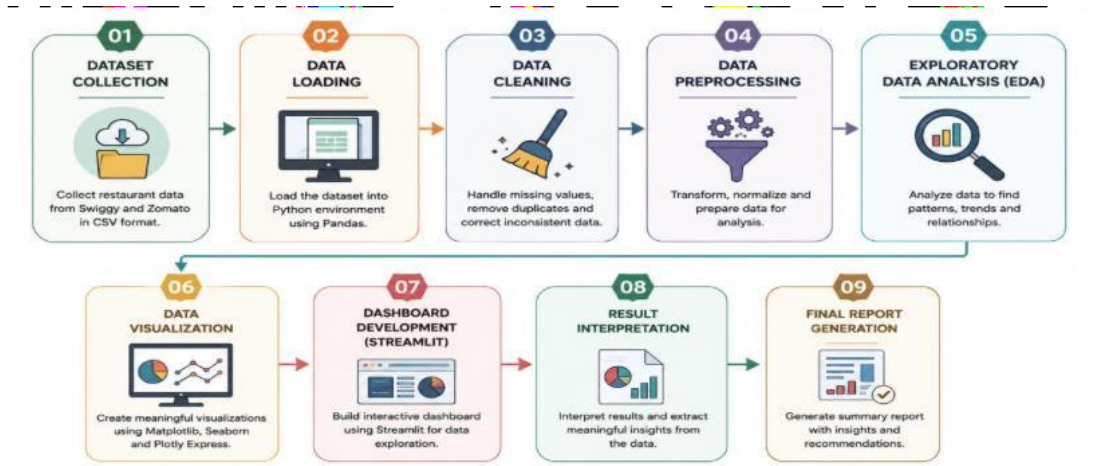


Figure 4.1: Data Flow Diagram

4.2 Flowchart

The flowchart represents the complete workflow of the proposed restaurant analytics system. The process begins with importing the CSV dataset. The system validates the dataset, performs preprocessing and cleaning, analyses restaurant information, generates visualizations, and finally displays the results through an interactive Streamlit dashboard.

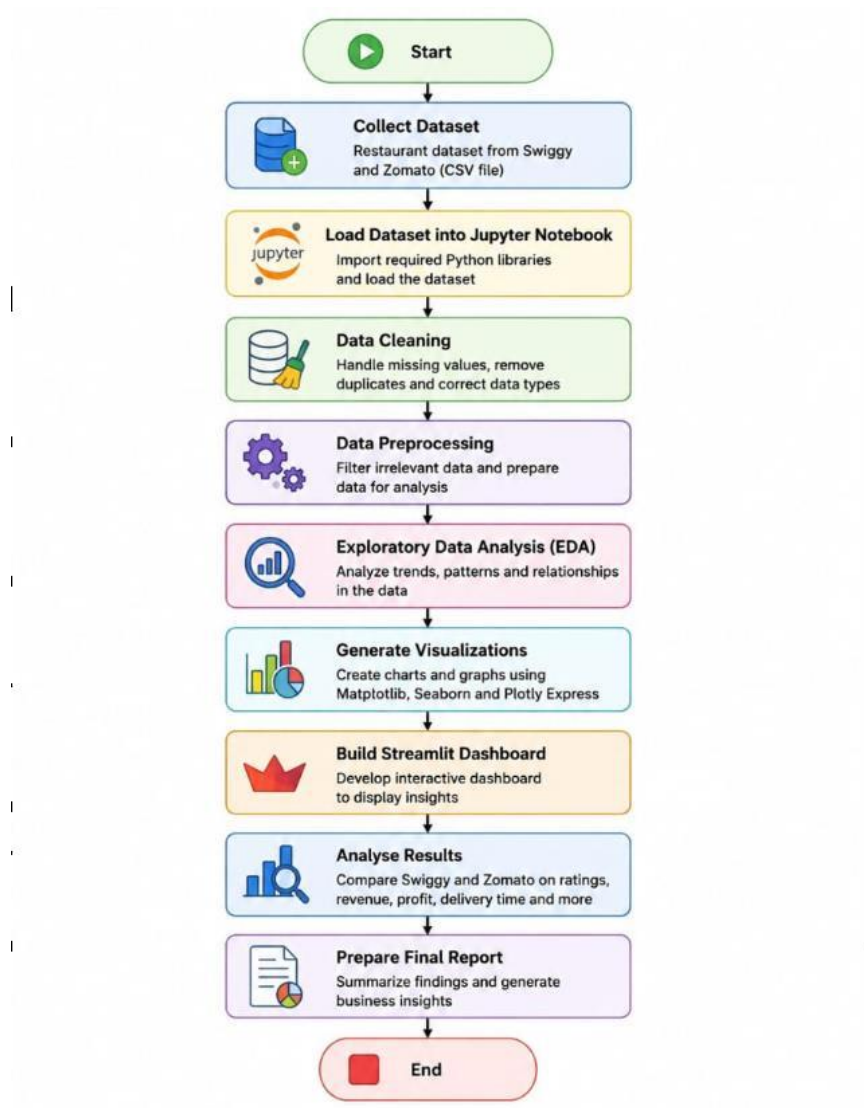


Figure 4.2: Project Flowchart

4.3 User Display and Report Format

The Streamlit dashboard provides a user-friendly interface for analysing restaurant data from Swiggy and Zomato.

Dashboard Features

- Home Page
- Dataset Preview
- Data Cleaning Summary
- Restaurant Comparison
- Ratings Analysis

- Revenue Analysis
- Delivery Performance
- Interactive Charts
- Sidebar Filters
- Summary Report

Report Format

The dashboard displays information in the following format:

- Dataset Overview
- Number of Restaurants
- Number of Features
- Missing Value Summary
- Statistical Summary
- Platform-wise Comparison
- City-wise Analysis
- Revenue Analysis
- Delivery Time Analysis
- Rating Analysis
- Customer Review Analysis
- Final Observations

4.4 Data Dictionary

The data dictionary describes the attributes available in the Swiggy vs Zomato dataset.

Table 4.1 Data Dictionary

Column Name	Description
Restaurant ID	Unique identification number of each restaurant.

Restaurant Name	Name of the restaurant.
City	City where the restaurant is located.
Locality	Area or locality of the restaurant.
Cuisine	Type of food served by the restaurant.
Restaurant Type	Category of restaurant (Cafe, Casual Dining, etc.).
Swiggy Rating	Rating received on Swiggy.
Zomato Rating	Rating received on Zomato.
Average Rating	Average rating across both platforms.
Swiggy Reviews	Total reviews on Swiggy.
Zomato Reviews	Total reviews on Zomato.
Total Reviews	Combined customer reviews.
Delivery Time	Estimated delivery time (minutes).
Delivery Fee	Delivery charges for customers.
Average Cost	Average meal cost for two people.
Monthly Orders	Estimated monthly orders received.
Monthly Revenue	Estimated monthly revenue generated.
Platform Commission	Commission charged by the platform.
Net Profit	Estimated restaurant profit.
Market Share	Restaurant market share percentage.
Better Performing Platform	Platform where restaurant performs better.
Amenities	Facilities provided by the restaurant.
Website/App Availability	Indicates online availability.
Food Licence	Food safety licence status.
Listing Date	Date when the restaurant was listed.

4.5 Functional Specifications

The major functionalities of the proposed system are:

- Import Swiggy vs Zomato dataset.
- Perform data cleaning and preprocessing.
- Remove duplicate records.
- Handle missing values.
- Generate descriptive statistics.
- Compare Swiggy and Zomato restaurant performance.
- Analyse ratings, reviews, delivery time, and revenue.
- Create interactive visualizations using Matplotlib, Seaborn, and Plotly Express.
- Display results through the Streamlit dashboard.
- Generate meaningful business insights.

4.6 Performance Specifications

The proposed system satisfies the following performance requirements:

- Fast dataset loading.
- Efficient preprocessing of restaurant data.
- Accurate statistical computations.
- Quick chart generation.
- Interactive dashboard response.
- Reliable comparison between Swiggy and Zomato.
- User-friendly interface.
- Accurate report generation.
- Scalable architecture for larger datasets.

CHAPTER 5

PROJECT PLAN

This chapter describes the development plan followed for the **Swiggy vs Zomato Restaurant Analytics Dashboard**. It outlines the project schedule and explains the programming languages, libraries, and software tools used to collect, preprocess, analyse, visualize, and present the restaurant data. A systematic development approach ensured the successful completion of the project.

5.1 Development Schedule

The development of the **Swiggy vs Zomato Restaurant Analytics Dashboard** followed a structured sequence of phases. Each phase focused on a specific task to ensure efficient project execution and timely completion.

Table 5.1: Development Schedule

Phase	Activity
Phase 1	Project planning and requirement analysis
Phase 2	Dataset collection and understanding
Phase 3	Data cleaning and preprocessing
Phase 4	Exploratory Data Analysis (EDA)
Phase 5	Data visualization and statistical analysis
Phase 6	Streamlit dashboard development
Phase 7	Testing and validation
Phase 8	Documentation and report preparation

5.2 Programming Languages and Development Tools

The **Swiggy vs Zomato Restaurant Analytics Dashboard** was developed using Python and several open-source libraries for data analysis, visualization, and dashboard development. These tools enabled efficient preprocessing, exploratory data analysis, business insights generation, and interactive visualization of restaurant data.

The programming languages and development tools used in this project are described below.

Python

Python is a high-level, interpreted programming language widely used in Data Science, Machine Learning, and Data Analytics because of its simplicity and extensive library support. In this project, Python served as the primary programming language for data cleaning, analysis, visualization, and dashboard development.



Figure 5.1: Python Logo

NumPy

NumPy is a numerical computing library that provides support for multidimensional arrays and mathematical operations. It was used to perform numerical calculations and assist in preprocessing the restaurant dataset.



Figure 5.2: NumPy Logo

Pandas

Pandas is a powerful data manipulation and analysis library. It was used to import the CSV dataset, clean missing values, remove duplicate records, transform data, and perform Exploratory Data Analysis (EDA).



Figure 5.3: Pandas Logo

Matplotlib

Matplotlib is a plotting library used for creating high-quality graphs and charts. It was used to visualize restaurant ratings, delivery time, monthly revenue, customer reviews, and other business metrics.



Figure 5.4: Matplotlib Logo

Seaborn

Seaborn is a statistical visualization library built on Matplotlib. It provides attractive and informative charts with minimal code. In this project, Seaborn was used to generate box plots, count plots, heatmaps, violin plots, and distribution plots for restaurant performance analysis.



Figure 5.5: Seaborn Logo

Plotly Express

Plotly Express is an interactive visualization library that allows users to explore data dynamically. It was used to create interactive bar charts, scatter plots, pie charts, line charts, and treemaps for comparing restaurant performance on Swiggy and Zomato.



Figure 5.6: Plotly Express Logo

Streamlit

Streamlit is an open-source Python framework used to build interactive data dashboards. In this project, Streamlit was used to create a user-friendly dashboard where users can upload the dataset, apply filters, compare restaurant statistics, and explore interactive visualizations.



Figure 5.7: Streamlit Logo

Jupyter Notebook

Jupyter Notebook is an interactive development environment that combines Python code, outputs, visualizations, and documentation in a single workspace. It was used during the development phase for data preprocessing, analysis, visualization, and testing.



Figure 5.8: Jupyter Notebook Logo

Microsoft Excel

Microsoft Excel was used for viewing and verifying the dataset before analysis. It helped inspect the structure of the CSV file, validate records, identify missing values, and perform basic data verification before importing the dataset into Python.



Figure 5.9: Microsoft Excel Logo

For your **Swiggy vs Zomato Restaurant Analytics Dashboard** project, **Chapter 6: Design** should explain the architecture and design of your system. Below is a professional version that you can directly include in your report.

6.1 Data Flow Diagram (DFD)

A **Data Flow Diagram (DFD)** is a graphical representation of how data moves through a system. It illustrates the input, processing, storage, and output of data within the application. The DFD helps in understanding the logical flow of information and the interaction between different system components.

The **Swiggy vs Zomato: Restaurant Performance and Customer Insights Analysis Dashboard** uses three levels of DFD to represent the overall workflow and internal processing of restaurant data.

6.1.1 Level 0 DFD (Context Diagram)

The Level 0 DFD, also known as the **Context Diagram**, provides a high-level overview of the entire system. It represents the dashboard as a single process interacting with the external user. The user uploads the restaurant dataset and selects filters, while the system processes the data and returns interactive dashboards, reports, and business insights.

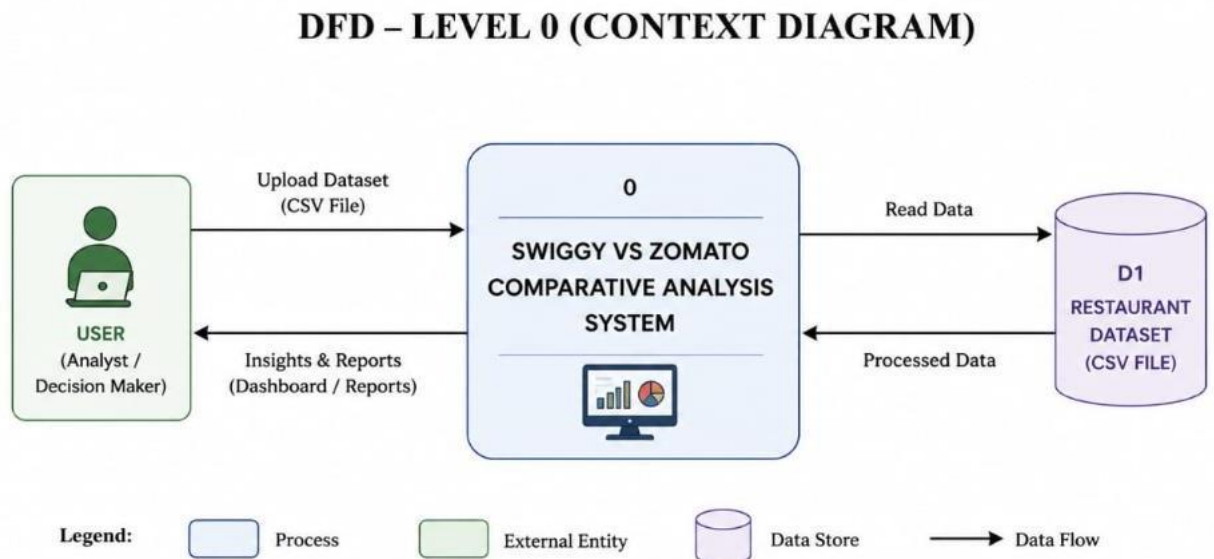


Figure 6.1: Context Level 0 Data Flow Diagram (DFD)

6.1.2 Level 1 DFD

The Level 1 DFD provides a detailed representation of the internal workflow of the system. It illustrates the major processing modules involved in data analysis.

The workflow begins by loading the restaurant dataset into the application. The dataset is then cleaned by handling missing values, duplicate records, and inconsistent data. After preprocessing, exploratory data analysis and statistical computations are performed. Finally, the processed information is displayed through an interactive Streamlit dashboard containing charts, KPIs, and reports.

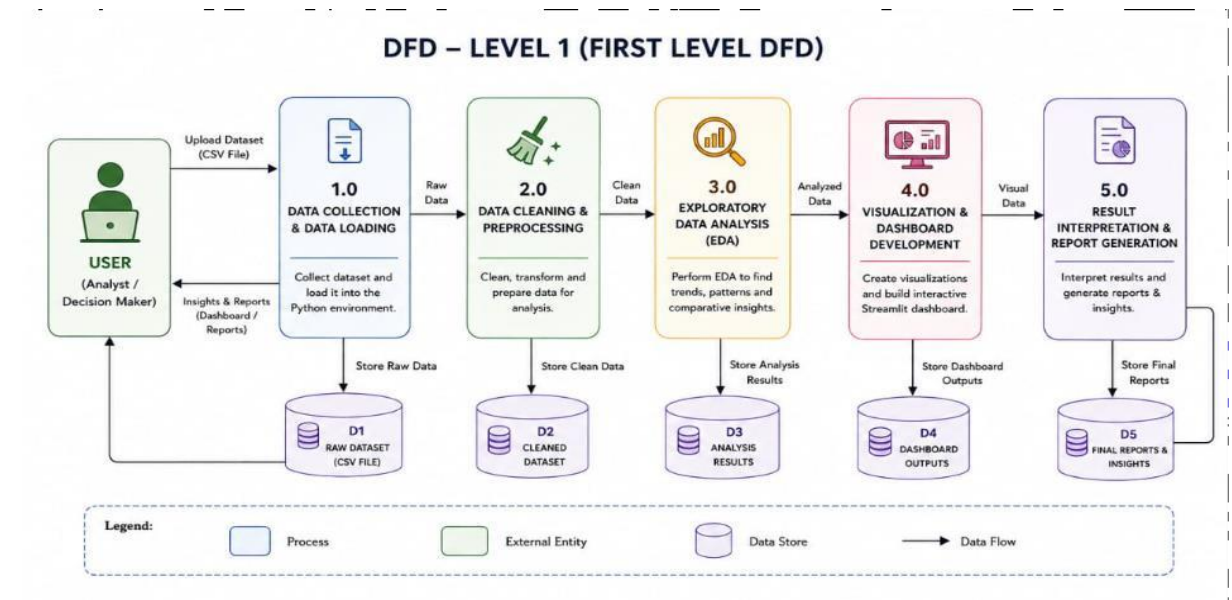


Figure 6.2: Level 1 Data Pipeline & Preprocessing DFD

6.1.3 Level 2 DFD

The Level 2 DFD further expands the data analysis process by showing how the cleaned dataset is transformed into meaningful business insights.

Initially, the user selects various filters such as city, cuisine, restaurant type, platform, and rating. Based on these selections, the system filters the dataset and calculates statistical measures including average ratings, revenue, delivery time, and restaurant counts. The processed data is then visualized using interactive charts created with Plotly and Matplotlib. Finally, the dashboard presents the results in an organized and user-friendly manner.

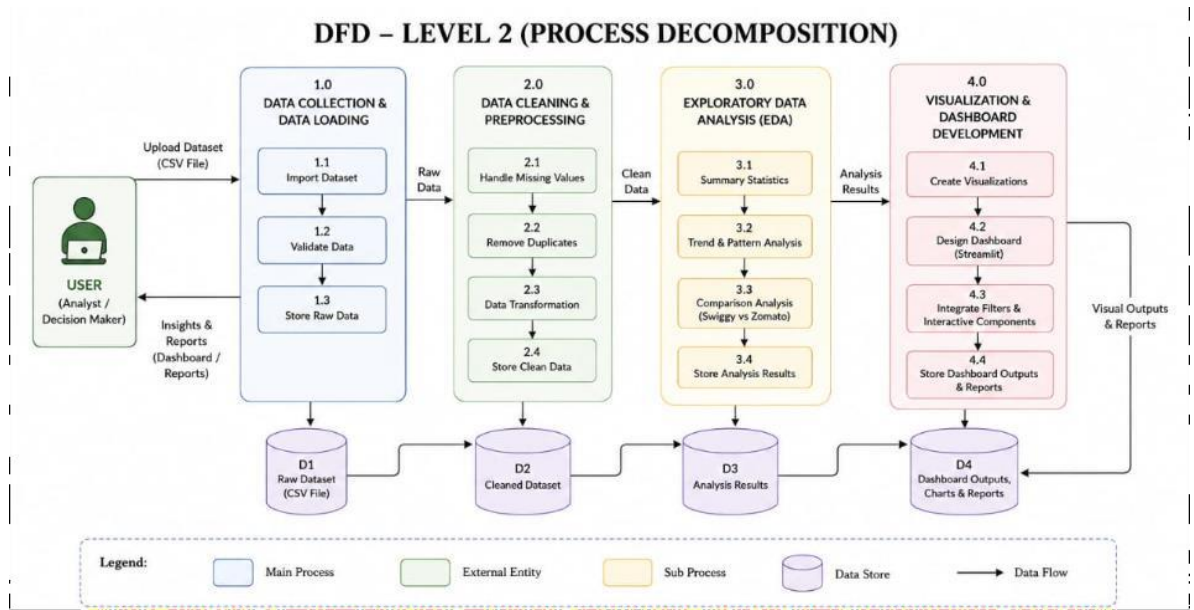


Figure 6.3: Level 2 Visualization Engine & Dashboard DFD

6.2 System Architecture

The system architecture describes the overall structure of the analytics dashboard and the interaction among different software components.

The application follows a simple data analytics pipeline consisting of five major stages:

1. Dataset Collection
2. Data Loading
3. Data Cleaning and Preprocessing
4. Data Analysis and Visualization
5. Interactive Dashboard Generation

The CSV dataset is loaded using the Pandas library. Data preprocessing techniques are applied to improve data quality by handling missing values, duplicates, and incorrect data formats. Exploratory Data Analysis (EDA) is then performed to discover trends, patterns, and relationships within the restaurant dataset.

The processed data is visualized using Plotly, Matplotlib, and Seaborn libraries. Finally, Streamlit is used to create an interactive dashboard where users can explore restaurant insights through filters, graphs, and key performance indicators.

6.3 Module Description

The application consists of the following functional modules:

6.3.1 Dataset Loading Module

This module imports the Swiggy vs Zomato restaurant dataset into the application using the Pandas library. It supports CSV file format and converts the data into a DataFrame for further processing.

6.3.2 Data Cleaning and Preprocessing Module

This module improves data quality by:

- Removing duplicate records
- Handling missing values
- Correcting inconsistent data
- Converting data types
- Formatting numerical values

The cleaned dataset becomes suitable for analysis.

6.3.3 Data Analysis Module

This module performs exploratory data analysis to generate meaningful insights from the restaurant dataset. Various statistical measures such as average ratings, revenue distribution, restaurant counts, delivery time, and cuisine popularity are calculated.

6.3.4 Visualization Module

This module creates interactive charts and graphs using Plotly, Matplotlib, and Seaborn. The visualizations help users understand restaurant performance and compare Swiggy and Zomato across different parameters.

6.3.5 Dashboard Module

The dashboard module integrates all analytical components into a user-friendly Streamlit interface. It allows users to interact with the data using filters and view dynamic visualizations, summary statistics, and business insights.

6.4 Input Design

The input to the system consists of the Swiggy vs Zomato restaurant dataset stored in CSV format. The dashboard also accepts user-defined filters such as:

- City
- Platform (Swiggy/Zomato)
- Cuisine

- Restaurant Type
- Rating
- Delivery Time

These inputs allow users to perform customized analysis.

6.5 Output Design

The output generated by the system includes:

- Interactive dashboard
- Bar charts
- Pie charts
- Scatter plots
- Line graphs
- Histograms
- Heatmaps
- Summary statistics
- Restaurant comparison reports
- Business insights

The outputs help users analyse restaurant performance and make data-driven decisions.

6.6 Database Tables

Table 6.1: Database Schema Structure & Attribute Data Types

Attribute Column	Data Type	Primary Constraint	Key Purpose
restaurant_id	VARCHAR(20)	PRIMARY KEY	Unique restaurant tag
restaurant_name	VARCHAR(10)	NOT NULL	Display name
city	VARCHAR(50)	INDEXED	Urban location

price_category	VARCHAR(30)	NOT NULL	Budget segmentation
swiggy_rating	FLOAT	CHECK (1.0 to 5.0)	Customer review score
zomato_rating	FLOAT	CHECK (1.0 to 5.0)	Customer review score
swiggy_revenue	BIGINT	NON-NEGATIVE	Monthly gross revenue
zomato_revenue	BIGINT	NON-NEGATIVE	Monthly gross revenue

CHAPTER 7

RESULTS AND DISCUSSION

Results:

The developed dashboard successfully analyzed and compared the performance of Swiggy and Zomato using a restaurant dataset containing 2,500 records and 41 attributes. Interactive visualizations enabled an effective comparison of restaurant ratings, delivery time, revenue, monthly orders, pricing, delivery charges, restaurant types, cuisines, and city-wise distribution. The analysis revealed that platform performance varies across restaurants, cities, and price categories, indicating that no single platform consistently outperforms the other. Restaurant location was found to influence delivery efficiency, as delivery time generally increased with greater distance from the city center. The dataset also showed that Budget and Mid-Range restaurants constitute the largest share of the restaurant listings, highlighting the dominance of affordable dining options.

Revenue and estimated monthly orders differed across cities and restaurant types, demonstrating that business performance is influenced by multiple operational factors. Furthermore, variations in delivery charges, commission percentages, and discount frequency were observed across restaurants. The interactive dashboard made these comparisons simple and provided meaningful business insights through clear visual representations.

Discussion:

The dashboard proved to be an effective tool for exploring restaurant performance across both food delivery platforms. Multiple visualizations helped identify trends in customer ratings, pricing, delivery efficiency, and revenue while allowing users to filter data by city, cuisine, and restaurant type.

The comparative analysis indicates that platform performance depends on a combination of restaurant characteristics rather than a single metric. Factors such as location, pricing category, restaurant type, delivery time, and customer ratings collectively influence business outcomes. These findings demonstrate the practical application of data analytics and interactive visualization in understanding restaurant performance and supporting informed decision-making.

7.1 Snapshot with Description



Fig 7.1: Dashboard Home Page

Description:

The home page of the dashboard provides an overview of the project. It contains the project title, sidebar filters, and navigation options. Users can select different categories and interact with the visualizations for detailed analysis.

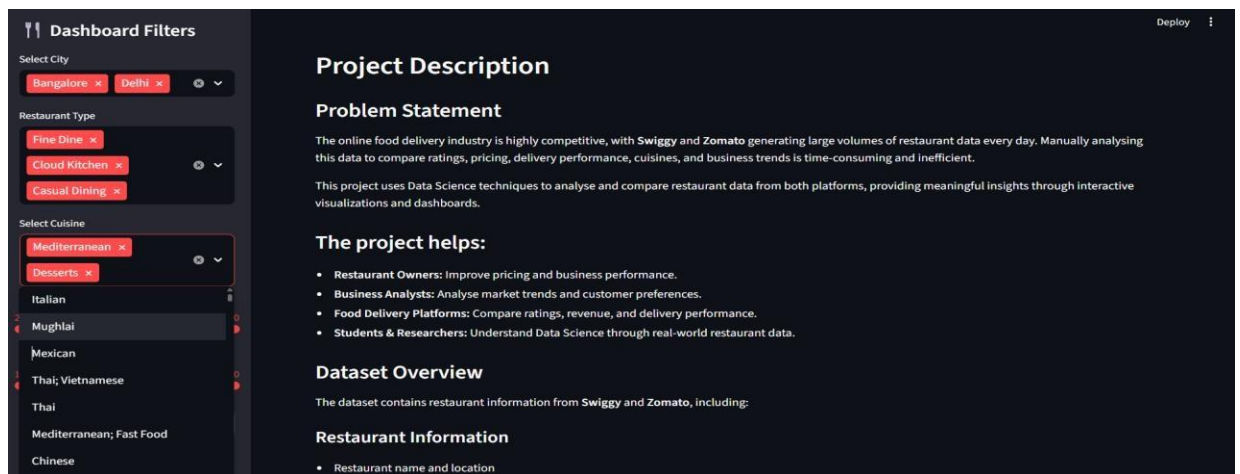


Figure 7.2: Project Description Dashboard

Description:

This figure shows the home page of the Streamlit dashboard. It includes interactive filters for city, restaurant type, and cuisine, along with the project description, problem statement, objectives, and dataset overview. The page provides users with an introduction to the project before viewing the analysis and visualizations.

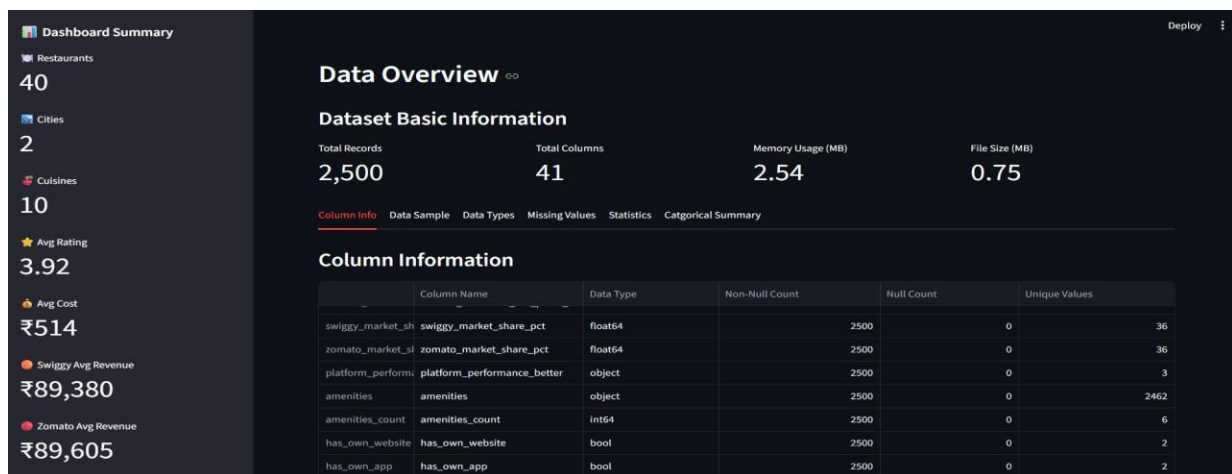


Figure 7.3: Data Overview Dashboard

Description:

This figure shows the Data Overview page of the Streamlit dashboard. It summarizes the dataset by displaying the total number of records, columns, memory usage, and file size. It also provides column information, data types, missing values, statistical summaries, and key dashboard metrics such as the number of restaurants, cities, cuisines, average rating, average cost, and estimated revenue.

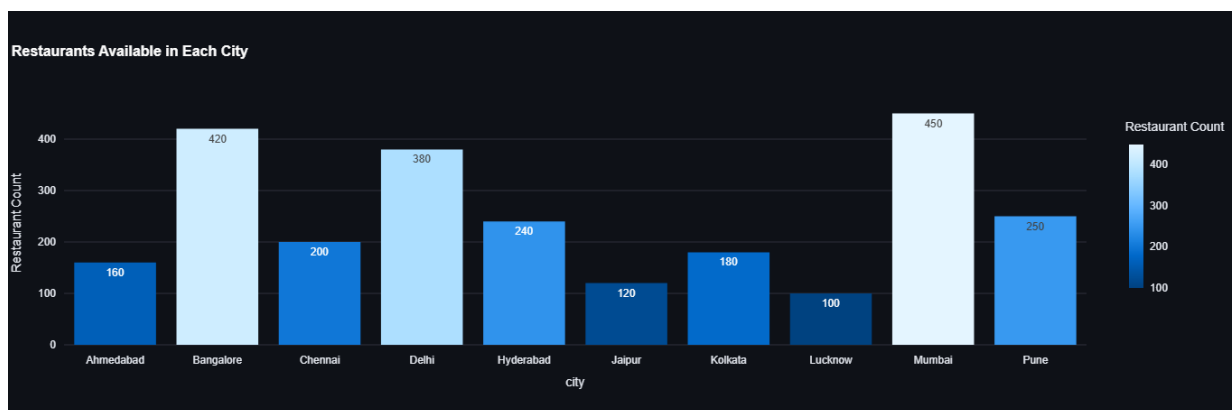


Figure 7.4: Restaurants by City-Bar Graph

Description:

Figure 7.4 shows the number of restaurants available in each city using a bar graph. It helps compare restaurant distribution across different cities and identify locations with the highest and lowest restaurant counts.

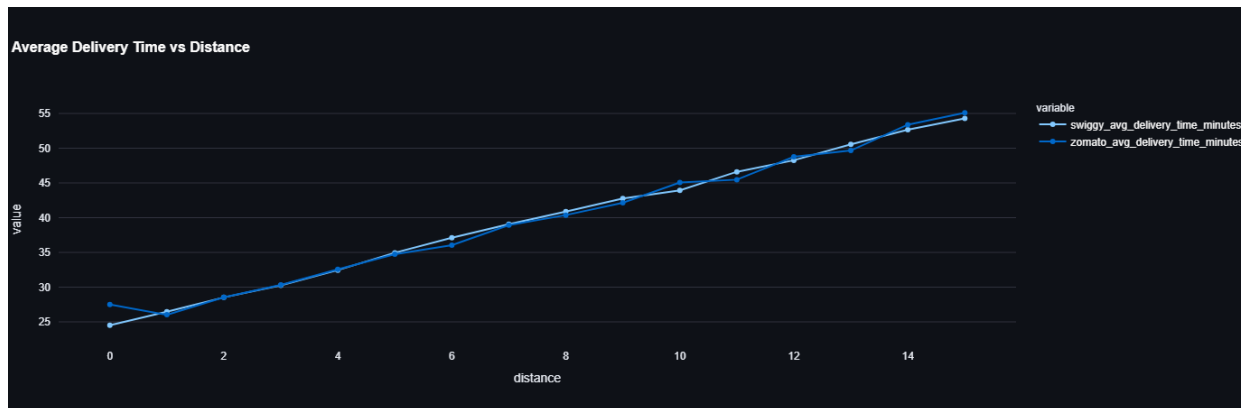


Figure 7.5: Average Delivery Time vs Distance-Line Chart

Description:

This line chart illustrates the relationship between the distance from the city center and the average delivery time of both Swiggy and Zomato. The graph shows that delivery time generally increases as the distance increases.



Figure 7.6: Platform Performance Distribution-Pie Chart

Description:

This pie chart displays the distribution of restaurants where Swiggy performs better, Zomato performs better, or both platforms perform equally. It helps compare the overall performance of both platforms across the dataset.

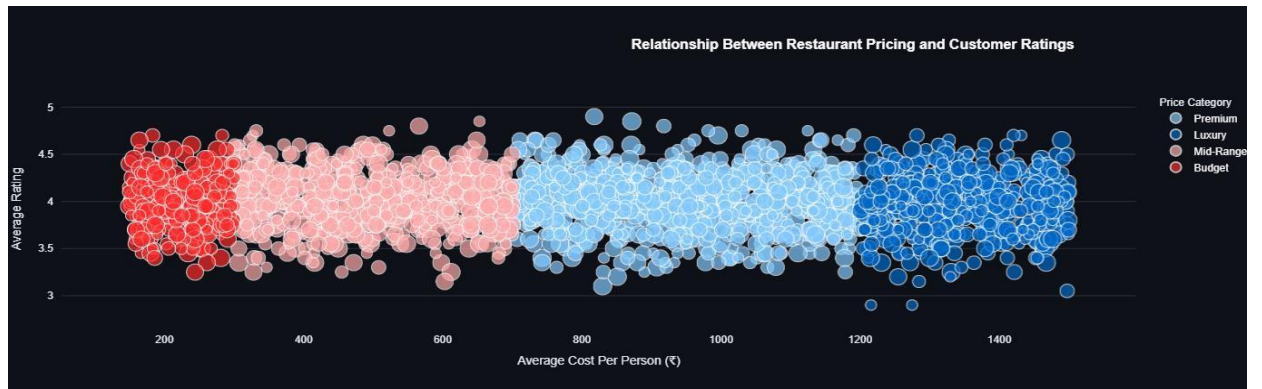


Figure 7.7: Relationship Between Restaurant Pricing and Customer Ratings – Scatter Plot

Description:

Figure 7.7 illustrates the relationship between the **average cost per person** and **customer ratings** across different price categories. It helps analyze whether higher-priced restaurants consistently receive higher customer ratings.

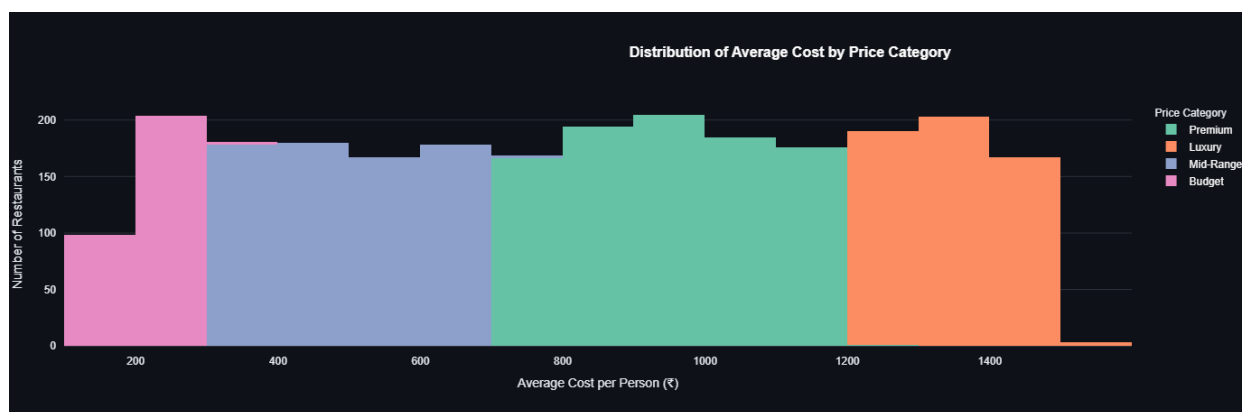


Figure 7.8: Distribution of Average Cost by Price Category – Histogram

Description:

Figure 7.8 illustrates the distribution of restaurants across different price categories based on the average cost per person. It helps compare the number of restaurants in each price range and understand the overall pricing distribution in the dataset.



Figure 7.9: Pricing Distribution Across Price Categories – Box Plot

Description:

Figure 7.9 presents the distribution of the **average cost per person** across different price categories and platform performance groups. It helps compare the spread, median, and variability of pricing among the restaurant categories.



Figure 7.10: Distribution of Restaurant Ratings by Better-Performing Platform – Violin Plot

Description:

Figure 7.10 illustrates the distribution of restaurant ratings across **Swiggy Better**, **Zomato Better**, and **Balanced** categories. It helps compare the spread, density, and median of ratings, providing insights into the performance of each platform.



Figure 7.11: Revenue Contribution by City and Restaurant Type – Treemap

Description:

Figure 7.11 presents the revenue contribution of different restaurant types across various cities using a treemap. It helps compare revenue distribution and identify the restaurant categories contributing the most revenue in each city.

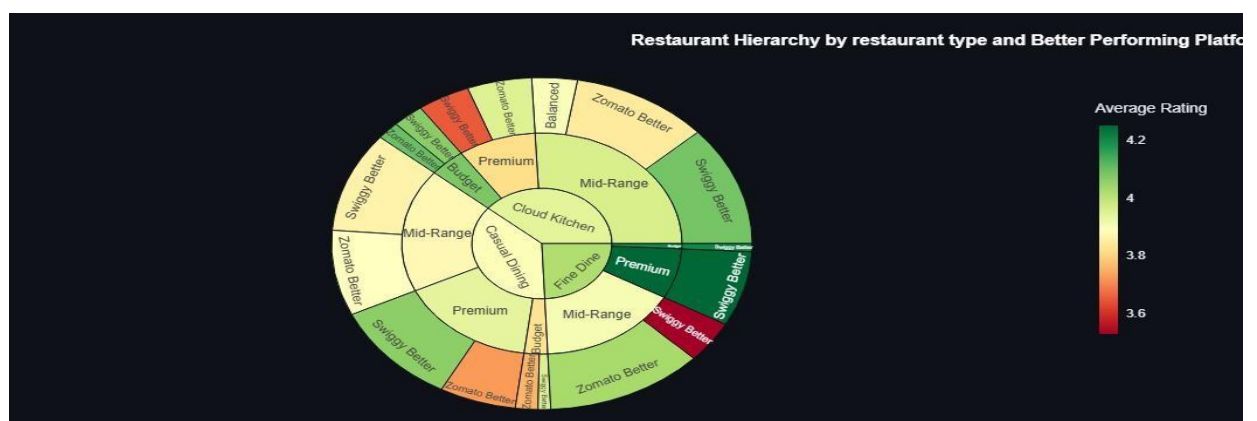


Figure 7.12: Restaurant Hierarchy by Restaurant Type and Better-Performing Platform – Sunburst Chart

Description:

Figure 7.12 presents the hierarchical distribution of restaurants based on **restaurant type, price category, and better-performing platform**. It helps visualize the relationship between these categories and compare platform performance across different restaurant segments.

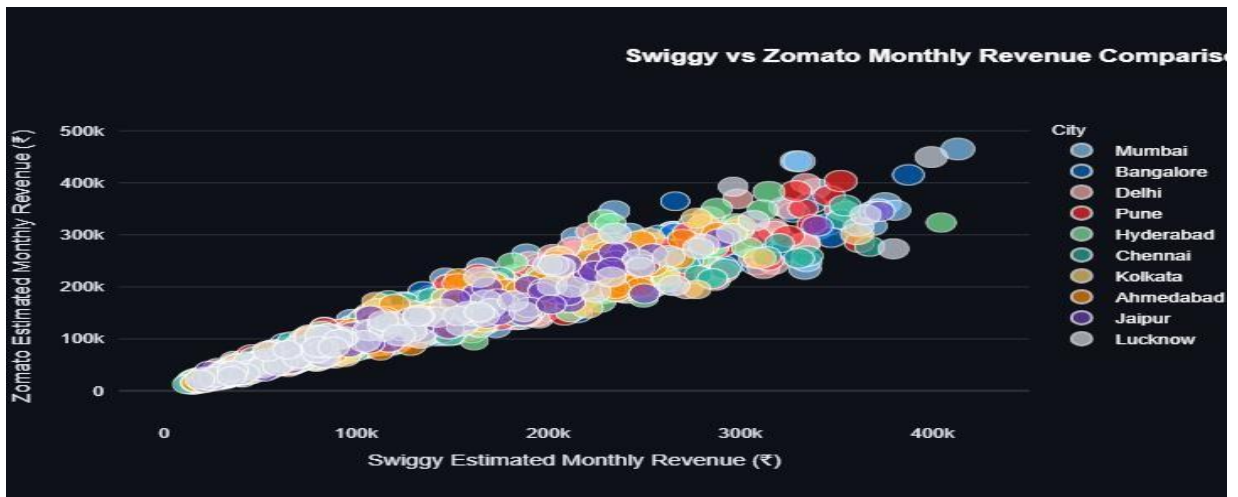


Figure 7.13: Swiggy vs Zomato Monthly Revenue Comparison – Bubble Chart

Description:

Figure 7.13 compares the estimated monthly revenue of Swiggy and Zomato across different cities using a scatter plot. It helps visualize the relationship between the revenues of both platforms and identify overall revenue trends and variations.

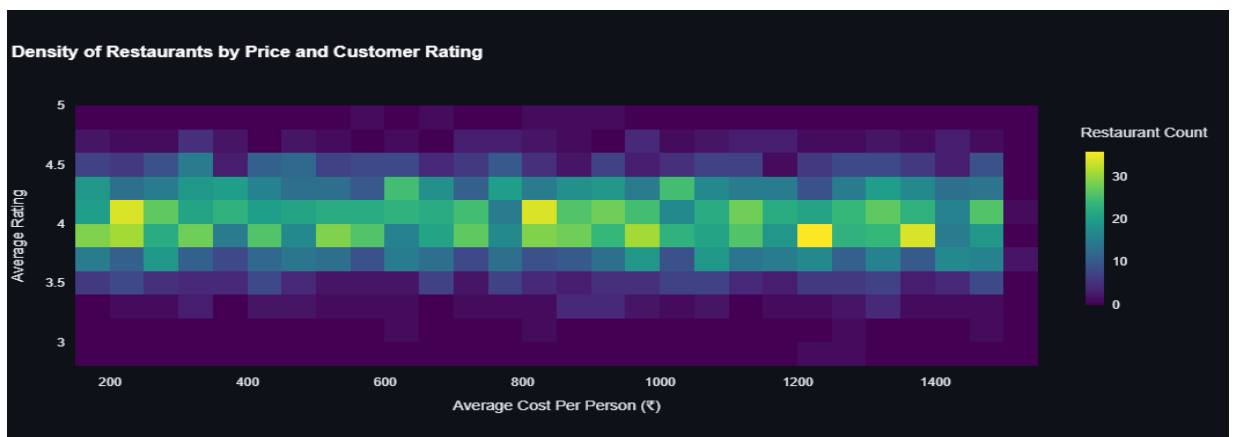


Figure 7.14: Density of Restaurants by Price and Customer Rating – Density Heatmap

Description:

Figure 7.14 illustrates the density of restaurants based on their average cost per person and customer ratings. It helps identify the most common combinations of price range and ratings, highlighting areas with the highest concentration of restaurants.

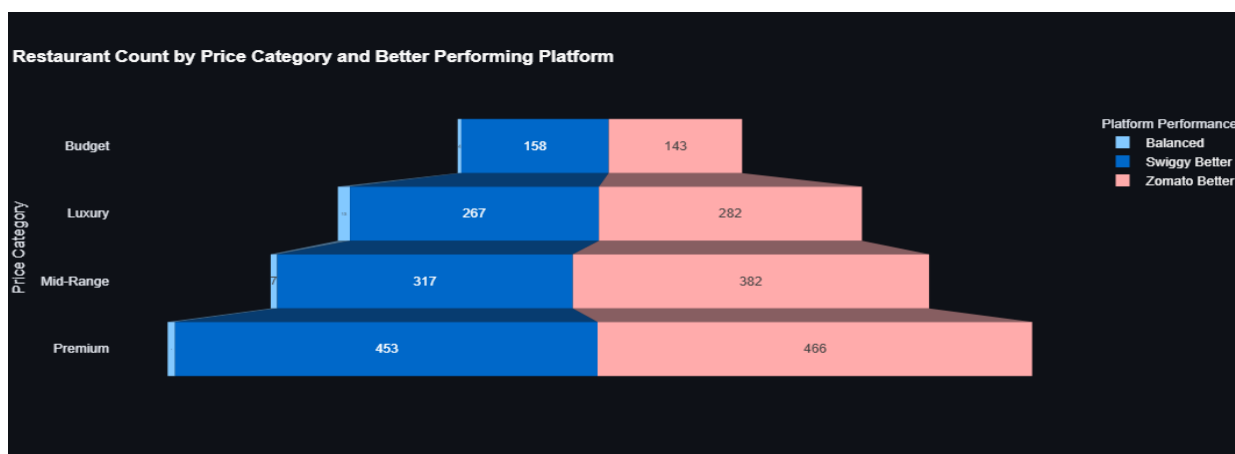


Figure 7.15: Restaurant Count by Price Category and Better-Performing Platform – Funnel Chart

Description:

Figure 7.15 presents the distribution of restaurants across different price categories based on the better-performing platform. It helps compare the number of restaurants where **Swiggy**, **Zomato**, or both platforms perform similarly within each price category.

7.1 Summary of Results

Table 7.1: Comparative Statistical Metrics Summary (Swiggy vs Zomato)

Operational Parameter	Swiggy Value	Zomato Value	Outperforming Platform
Total Restaurants Analyzed	2,500	2,500	Shared Base
Average Rating Score (out of 5.0)	3.99	4.01	Zomato (+0.02)
Average Delivery Speed (Minutes)	39.67 mins	39.57 mins	Zomato (0.10 mins faster)
Total Monthly Revenue Generated	₹32,45,15,484	₹32,56,10,971	Zomato(+₹10.95 Lakhs)

Average Monthly Net Profit / Rest.	₹1,24,030.65	₹1,24,435.13	Zomato (+₹404.48)
Outperforming Restaurant Count	1,195 (47.80%)	1,273 (50.92%)	Zomato (+78 Listings)
Balanced Performance Count	—	—	32 Listings (1.28%)

CONCLUSIONS AND FUTURE SCOPE

8.1 Conclusion

The **Swiggy vs Zomato: Restaurant Performance and Customer Insights Analysis** project successfully designed and built a comparative data analytics dashboard using **Python, Pandas, Plotly**, and **Streamlit**. By processing **2,500 restaurant listings** across **10 major Indian cities**, the application provides clear insights into platform ratings, delivery times, commission costs, and net revenues. The empirical findings show close competition, with Zomato holding a slight edge in total revenue, average ratings, and performance dominance across the dataset.

8.2 Future Scope

- **Machine Learning Integration:** Building predictive regression models (e.g., XGBoost, Random Forest) to forecast monthly restaurant revenues based on delivery distance, discount frequency, and price category.
- **Sentiment Analysis:** Utilizing Natural Language Processing (NLP) models to analyze customer review text and identify operational improvement areas.
- **Real-time API Ingestion:** Connecting the Streamlit application to live food aggregator APIs for real-time tracking of menu pricing and delivery changes.

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